When One Character Erases a Word: Orthographic Noise, Tokenizers and Negation in Arabic Sentiment Classification

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Abstract

Arabic tweets are full of orthographic variation that readers barely notice: the elongation character tatweel, emphatic letter repetition, and inconsistent spelling of hamza and final letters. We measure how this variation affects five Arabic and multilingual encoders fine-tuned for three-way sentiment classification on two tweet corpora, ArSarcasm-v2 and ArSAS. The study uses three training seeds per model and about 100,000 controlled edits per model, and was preregistered and frozen before a single evaluation on the held-out test sets. At the input level the effect is stark: in all four Arabic WordPiece models, one tatweel turns the edited word into an unknown token, including every edited negation word. Predictions are sturdier than this suggests. On the test sets, a tatweel in one word changes 3–9% of predictions, and in 30% of the words 10–22%. How badly a tokenizer damages a word predicts these rates only weakly and inconsistently. It matters more for what the changes cost: under dense tatweel the Arabic models lose 3.7–5.6 Macro-F1 points, against 1.4 and 2.7 for XLM-R. Two annotators judged every sampled tatweel edit to preserve meaning and sentiment, so these changes are errors. Editing the negation word itself adds only a small and inconsistent effect. Removing tatweel and collapsing letter runs before tokenization undoes the edits at no measurable cost on clean text. On test tweets that already contain letter runs, it raises Macro-F1 by 4–8 points on ArSarcasm-v2. A negation-aware normalizer we had proposed earlier adds nothing. We release the perturbation suite, leakage-safe splits, predictions and the preregistration.

1 Introduction

Social media Arabic is written with considerable freedom. Writers stretch a word with the tatweel character for visual effect (جميـــل for جميل "beautiful"), repeat a letter for emphasis (كثيييير, حلووو), and often omit hamza or write final ة as ه. In the two corpora we study, 3–5% of training and validation tweets contain a tatweel, and a similar share repeat a letter three or more times (Table 7). Native readers glide over these variants. Whether sentiment classifiers do so is less clear, because the tokenizers that feed them were built from vocabularies in which these forms are rare or absent.

We arrived at this question through a negative result. A negation-aware normalization method we developed earlier did no better than a simpler normalizer under any condition. In that experiment, single-word edits applied to one MARBERT model produced accuracy changes too small to separate from zero. We concluded that the design, not the models, made the experiment uninformative. A single model gives no basis for comparison. A single edited word per tweet is a weak treatment. The repetition edits also skipped the long vowels ا, و and ي, which carry most emphatic repetition in real tweets.

This paper replaces that experiment with a broader, preregistered audit. We fine-tune five encoders with three seeds each on two tweet datasets. We then apply four families of orthographic edits at two densities, and in some conditions we edit the negation word itself. Two annotators, and a third for their disagreements, judged whether the edits preserve meaning and sentiment. After development on the validation sets, we froze the study and evaluated it once on the test sets. We ask four questions: (RQ1) how often do realistic edits change predictions; (RQ2) do tokenizer effects explain differences between models; (RQ3) is editing the negation word worse than editing another word; and (RQ4) does simple normalization before tokenization undo the damage?

The main findings, all of which hold on the held-out test sets, are:

- Arabic WordPiece tokenizers map any word containing tatweel to `[UNK]`. This includes

negation words, so an edited tweet such as لا reaches the model with its negation removed (§4).

- Predictions change far less than this input damage would suggest. Tatweel in one word changes 3–9% of them, and tatweel in 30% of the words changes 10–22%. The Arabic models' changes are costlier, however: on ArSarcasm-v2 each point of flip rate costs them about 2.5 times as much Macro-F1 as it costs XLM-R (§6).

- Editing the negation word adds at most 2.5 points of flip rate over editing another word, and the difference is significant in only half of the test comparisons (§7).

- Stripping tatweel and collapsing letter runs removes nearly all of these changes at no measurable cost on clean text. On test tweets with natural letter runs it also improves accuracy. The negation guard of our earlier method adds nothing (§§8–9).

2 Related work

Noise and robustness. Both synthetic and natural character noise degrade neural models (Belinkov and Bisk, 2018). Character-level recognizers have been proposed to defend against misspellings (Pruthi et al., 2019), and behavioural test suites probe whether a model is invariant to such edits (Ribeiro et al., 2020). Our edits are specific to Arabic script. We separate edits presumed to preserve meaning from a typo control and verify that separation with human judgments. We also report the costs of a defense alongside its benefits.

Arabic pretrained models. MARBERT (Abdul-Mageed et al., 2021), AraBERT (Antoun et al., 2020) and CAMeLBERT (Inoue et al., 2021) use WordPiece vocabularies learned from Arabic text. MARBERT and its successor share a vocabulary built from tweets. XLM-R (Conneau et al., 2020) uses a multilingual SentencePiece vocabulary that falls back on individual characters and therefore never produces an unknown token. The recommended preprocessing for some Arabic models removes tatweel, but nothing in the model interface enforces this. We are not aware of work that measures what omitting that step costs a fine-tuned model.

Family	Original	Edited
Tatweel	حلو	حلـو
Elongation ×2	حلو	حلووو
Elongation ×4	كثير	كثيييير
Folding	جميلة / أحب	جميله / احب
Typo (control)	جيد	جبد
Cue (tatweel)	لا	لـا

Table 1: The edit families, illustrated on single words.

Negation. Benchmarks under-represent negation, and models trained on them often ignore it (Hossain et al., 2020). We do not parse negation scope. Instead, we compare an edit on the negation word with the same edit on another word in the same tweet.

3 Study design

Protocol. We preregistered the conditions, metrics, hypotheses and decision rules before observing any model prediction on the new edits, and we developed the analysis on the validation sets. A written amendment fixed the test protocol. The study was then frozen at a tagged commit, and only after that were the test sets perturbed and evaluated, once. We report the test sets as the main results and give the validation results in Appendix C. The tokenizer audit in §4 was run before the preregistration and is disclosed there.

Data. ArSarcasm-v2 (Abu Farha et al., 2021) contains dialectal and MSA tweets labelled negative, neutral or positive. We use a leakage-safe split that removes from training any tweet whose normalized text occurs in the official test set. This gives 10,037 training, 2,509 validation and 3,000 official test tweets. ArSAS (Elmadany et al., 2018) contains 21,064 tweets with an additional Mixed class, which we drop. We merge repeated tweet IDs and identical texts into groups, then exclude the 90 tweets whose groups carry conflicting labels and 44 further copies linked to excluded tweets. A stratified group split then gives 13,738 training, 1,966 validation and 3,923 test tweets. We considered ASTD (Nabil et al., 2015) as the second dataset and rejected it, because 20% of ArSarcasm-v2 training tweets (2,473) also occur in ASTD. ArSAS shares one tweet with ArSarcasm-v2, and that tweet is excluded. Each model is trained and evaluated separately on each dataset.

Edits. Table 1 illustrates the four families.

- *Tatweel* inserts one ـ after the last letter that joins to the next.
- *Elongation* repeats the last internal long vowel, or the final letter if there is none, two or four extra times.
- *Folding* writes أ, إ and آ as ا, final ة as ه, and final ى as ي.
- *Typo* replaces one interior letter with its neighbour on the Arabic keyboard.

The typo family was intended as a control that damages meaning. The human judgments in §5 show that it does so only partly.

Each family is applied at two densities: *single* edits one word, and *dense* edits $\lceil 0.3n \rceil$ of the n Arabic words in the tweet. In *cue* conditions the edit falls on an unambiguous negation word (لا, لم, لن, ليس, مش, مو, مافي, بدون and their variants). We leave out ما, whose relative, interrogative and exclamative uses outnumber its negating use. Non-cue conditions never edit a negation word. They also skip URLs, mentions, hashtags and tokens attached to Latin characters. Which word gets edited is decided by hashing the tweet ID and the condition. Single edits are a subset of the dense edits for the same tweet, and both elongation levels edit the same character.

On the test sets, edits apply to between 2,119 and 2,950 of the 3,000 ArSarcasm-v2 tweets per condition, and to between 3,303 and 3,898 of the 3,923 ArSAS tweets. Cue conditions apply to the 626 and 623 test tweets that contain an editable negation word. Across both partitions this amounts to about 100,000 edited tweets per model.

Models. We fine-tune MARBERT, MARBERTv2, AraBERTv02-twitter, CAMeLBERT-DA and XLM-R-base with one fixed recipe:

- AdamW, learning rate 2×10^{-5}, batch size 16, at most five epochs;
- early stopping and checkpoint selection on validation Macro-F1;
- seeds 42, 43 and 44.

This gives 30 models. Revisions of all pretrained weights are pinned. Table 2 reports clean performance. ArSarcasm-v2's official test set is markedly harder than its validation split, and every model scores 8–12 points lower on it.

(a) Validation, mean ± s.d. over seeds

Model	ArSarcasm-v2	ArSAS
MARBERT	74.7 ± 0.3	83.1 ± 0.3
MARBERTv2	75.8 ± 0.3	83.5 ± 0.2
AraBERT-tw	74.4 ± 0.2	83.3 ± 0.3
CAMeL-DA	71.0 ± 0.3	81.4 ± 0.3
XLM-R	71.0 ± 0.4	80.5 ± 0.2

(b) Test, 95% interval

Model	ArSarcasm-v2	ArSAS
MARBERT	66.5 [64.8, 68.1]	83.1 [82.0, 84.2]
MARBERTv2	65.8 [64.2, 67.4]	83.7 [82.7, 84.7]
AraBERT-tw	64.9 [63.2, 66.6]	83.9 [82.8, 84.9]
CAMeL-DA	62.7 [61.0, 64.3]	81.9 [80.8, 83.1]
XLM-R	59.3 [57.6, 61.0]	81.9 [80.9, 83.1]

Table 2: Clean Macro-F1 (%). The test figures are seed-averaged and computed on the 2,948 and 3,898 test tweets eligible for single-word tatweel.

Defenses. At inference time we compare preprocessors, each applied identically to clean and edited text:

- RAW leaves the input unchanged;
- STD removes tatweel and collapses runs of three or more identical letters to a single letter;
- NO_CTX and NASN2 are the two conservative normalizers from our earlier work.

NO_CTX and NASN2 share all of their rules except one: NASN2 also declines to collapse repeated letters near a negation word. Their repetition rule covers consonants only, so they do not undo elongated long vowels. The comparison between these two was settled on validation (H5 below). The test protocol therefore evaluates RAW and STD only.

Metrics and statistics. The primary metric is the *flip rate*, the share of edited tweets whose predicted class differs from the prediction for the clean tweet. It needs no gold label and isolates the effect of the edit. We also report Macro-F1 on clean and edited versions of the same tweets and the difference between the two.

Intervals are 95% percentiles of 2,000 cluster-bootstrap resamples. Clusters are groups of duplicate texts, and the rate for each resample is averaged over the three seeds. Contrasts are paired on identical tweets. These include cue versus non-cue, dense versus single, a defense versus RAW, and each WordPiece model versus XLM-R. A hypothesis counts as supported for a model when its interval excludes zero in the stated direction.

Field	Agreement	κ	AC1
Meaning preserved	89.0%	0.43	0.88
Sentiment preserved	83.9%	0.56	0.80
Sentiment of edited text	80.3%	0.69	0.71
Negation present	92.9%	0.85	0.87
Negation affected by edit	91.3%	0.84	0.88

Table 3: Agreement between the two annotators on 310 items (20 pilot items excluded).

Deviation from the preregistration. The preregistered zero-shot LLM was not evaluated, because the required GPU was unavailable.

4 What the tokenizer sees

Figure 1 shows, for every model and edit family, how many tokens an edited word receives relative to the original. The pattern is the same on both datasets and both partitions.

None of the four Arabic WordPiece vocabularies contains the tatweel character. WordPiece therefore replaces any word containing it with a single `[UNK]`. This happens in 100% of tatweel edits, including every edited negation word (1,969 across the four partitions). A tweet containing لا thus reaches these models with no negation. XLM-R never produces an unknown token. It instead splits a word containing tatweel into 2.3–2.6 times as many pieces, and an edited negation word into 2.9 times as many. Elongation fragments the word in every model: four extra letters multiply its token count by 2.4–4.1. Folding is largely absorbed by MARBERT, whose lowercasing strips hamza from alef. As a result, 31–39% of its single-word folding edits leave the token sequence unchanged. The other models see nearly every folding edit.

5 Do humans notice the edits?

The study rests on the claim that most of these edits preserve meaning. To test it, two native-speaker annotators independently judged 330 edited tweets drawn from the ArSarcasm-v2 validation set, 30 per condition and one edit per tweet. They answered five questions per item. Their first 20 items served as a pilot and are excluded from the agreement figures. Agreement (Table 3) is high on every field. Cohen's κ is low for meaning preservation because almost every answer is "yes". Gwet's AC1 (Gwet, 2008) corrects for this skew and gives 0.88.

A third annotator resolved the 87 items on which the two disagreed about meaning or sentiment preservation. That annotator saw both earlier answers, without the annotators' identities or notes and in random order. On meaning, the third annotator sided with each of the two about equally often (76% and 75%). On sentiment, the third annotator sided more often with annotator A (52%) than with annotator B (21%), who had marked many of these items unsure.

The results (Table 4) separate the families clearly.

- *Tatweel* preserved both meaning and sentiment in all 90 items, including the 30 in which it fell on a negation word. Any prediction change under tatweel is therefore an error.
- *Elongation* preserved meaning in all 120 items, but it often changed sentiment. The sentiment was judged unchanged in only 17 and 14 of 30 items for one-word elongation by two and four letters, in 6 of 30 when 30% of the words were elongated, and in none of the 30 items with an elongated negation word. Readers take repetition as emphasis, so a prediction change under elongation is not necessarily an error.
- *Folding* preserved meaning in 48 of 60 items by agreement, and in all 60 after adjudication. The disputed items wrote على as علي or dropped a hamza, which the third annotator judged cosmetic.
- *Typos*, meant as the control, were often read through. After adjudication, 19 of 30 single-word typos and 8 of 30 dense ones preserved meaning and sentiment.

Adjudication changed none of the counts for tatweel or elongation. Every elongation item it resolved was judged meaning-preserving but sentiment-changing.

6 How often do predictions change?

Figure 2 shows the test flip rates. A single edited word rarely matters.

- Tatweel in one word changes 7.0–9.3% of predictions on ArSarcasm-v2 and 3.0–3.9% on ArSAS.
- Two extra letters change 2.9–7.8%.
- Folding changes 0.9–4.3%.

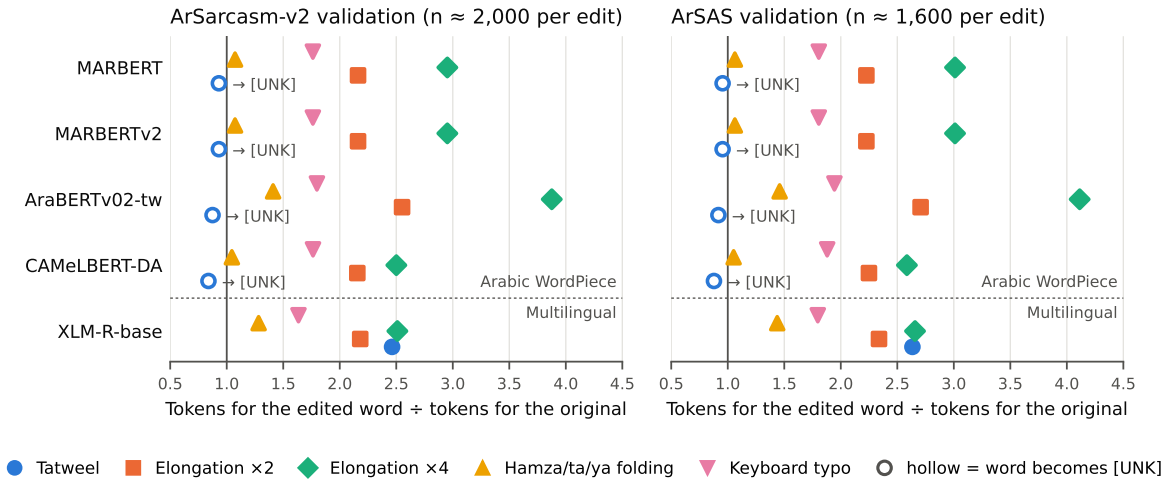


Figure 1: Number of tokens produced for an edited word, relative to the original word, averaged over single-word edits on the validation sets. A hollow marker means the edited word becomes `[UNK]` (rate 1.00 in every such cell). The test sets give the same pattern. MARBERT and MARBERTv2 share a vocabulary, so their rows coincide.

Condition	Meaning (A / B)	Both: meaning	Both: meaning + sentiment	Final
Tatweel, 1 word	30 / 30	30	30	30
Tatweel, 30% of words	30 / 30	30	30	30
Tatweel on negation cue	30 / 30	30	30	30
Elongation ×2, 1 word	30 / 30	30	17	17
Elongation ×4, 1 word	30 / 30	30	14	14
Elongation ×2, 30% of words	30 / 30	30	6	6
Elongation ×2 on negation cue	30 / 30	30	0	0
Folding, 1 word	30 / 25	25	25	30
Folding, 30% of words	29 / 24	23	23	30
Typo (control), 1 word	24 / 20	18	18	19
Typo (control), 30% of words	13 / 11	8	8	8

Table 4: Items, out of 30 per condition, judged meaning-preserving by annotator A and by annotator B; items for which both judged meaning preserved, or both meaning and sentiment preserved; and the final count of meaning- and sentiment-preserving items after a third annotator resolved the disagreements.

Density matters a great deal. When 30% of the words carry tatweel, flip rates rise to 13.4–21.6% on ArSarcasm-v2 and 9.6–11.3% on ArSAS. Dense elongation reaches 16.2% for XLM-R and 15.3% for CAMeLBERT-DA. Every dense-versus-single contrast for the three presumed-preserving families is positive, in all five models and on both datasets (30 of 30 on test, as on validation; H1). The increases range from 1.1 to 3.8 points for folding and from 6.1 to 12.2 points for tatweel and elongation. Tatweel changes more predictions on the ArSarcasm-v2 test set than on its validation split, in line with the lower clean accuracy, while elongation and typos change slightly fewer.

Tokenizer damage barely predicts flip rates. The tokenizer audit predicts a large gap between WordPiece models, which lose the edited word entirely, and XLM-R, which keeps a fragmented version. The gap exists but depends on the dataset. Under dense tatweel on ArSarcasm-v2, every WordPiece model flips more often than XLM-R, by 3.8–8.1 points. On ArSAS three of the four do so, by 1.4–1.8 points (H2: 7 of 8 test contrasts, 6 of 8 on validation). On ArSarcasm-v2, XLM-R is the most sensitive model to elongation.

Across the 55 model–condition cells per dataset, the rank correlation between the `[UNK]` rate of the edited words and the flip rate is 0.44 on ArSarcasm-v2 test but only 0.07 on ArSAS test. On validation it is 0.18 and 0.14. Fragmentation does not correlate with flip rate on any partition ($|\rho| \leq 0.11$). One explanation is that fine-tuning already exposes the models to `[UNK]` for tatweel words, since 4.6% of ArSarcasm-v2 training and validation tweets contain a natural tatweel.

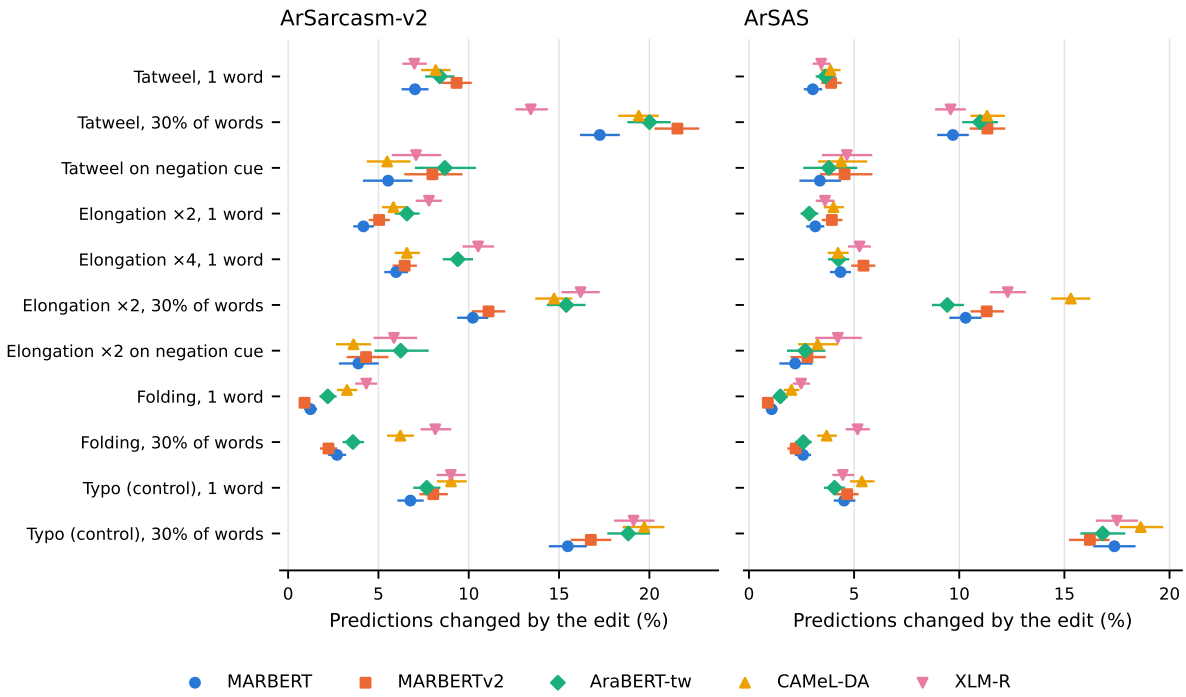


Figure 2: Test-set flip rates with 95% cluster-bootstrap intervals (RAW inputs, three seeds averaged per model). Full tables for both partitions are in the appendix.

	ArSarcasm-v2			ArSAS		
	Tatweel	Elong. ×2	Typo	Tatweel	Elong. ×2	Typo
MARBERT	4.9 [3.7, 6.0]	3.5 [2.6, 4.5]	4.6 [3.3, 5.7]	3.8 [3.0, 4.6]	4.0 [3.3, 4.8]	11.4 [10.4, 12.5]
MARBERTv2	5.5 [4.1, 6.7]	3.7 [2.8, 4.7]	3.3 [2.1, 4.5]	3.7 [2.9, 4.5]	3.9 [3.1, 4.7]	7.1 [6.1, 8.1]
AraBERT-tw	5.6 [4.3, 6.8]	4.6 [3.5, 5.7]	4.6 [3.2, 6.0]	3.8 [3.0, 4.6]	3.0 [2.2, 3.7]	9.9 [8.9, 11.0]
CAMeL-DA	5.1 [3.9, 6.3]	3.1 [2.0, 4.2]	4.3 [2.8, 5.6]	4.0 [3.1, 4.9]	6.1 [5.1, 7.0]	9.5 [8.4, 10.6]
XLM-R	1.4 [0.4, 2.5]	4.5 [3.3, 5.7]	4.3 [2.9, 5.7]	2.7 [2.1, 3.4]	3.6 [2.8, 4.4]	7.6 [6.6, 8.6]

Table 5: Test-set Macro-F1 drop in points (clean minus edited, same tweets) under dense edits, with 95% intervals.

But WordPiece errors cost more. The flip rate does not say whether a changed prediction is wrong. Table 5 does. Under dense tatweel on the test sets, Macro-F1 falls by 4.9–5.6 points for the WordPiece models on ArSarcasm-v2 and by 3.7–4.0 points on ArSAS. For XLM-R it falls by 1.4 and 2.7 points. Per point of flip rate, the WordPiece models lose 0.25–0.28 Macro-F1 points on ArSarcasm-v2, against 0.10 for XLM-R. On ArSAS the gap is smaller: 0.33–0.39 against 0.28. Validation shows the same ordering. This is consistent with an erased sentiment word removing the evidence for the label rather than merely adding noise. Dense typos cost the most on ArSAS (7.1–11.4 points).

7 Negation as a stress test

Natural noise on a negation word is rare. In the training and validation tweets, at most five per corpus contain an elongated or tatweel-marked negation word. We therefore treat the cue conditions as a controlled probe of how models represent negation, not as an estimate of real-world risk.

For each tweet with an editable negation word, we compare editing that word with editing another word in the same tweet (H3). Tatweel on the negation word, which turns it into `[UNK]`, always changes at least as many predictions as tatweel elsewhere. The difference is small, however. On the test sets the increase is 0.3–2.5 points, and it is significant in four of the eight WordPiece model–dataset pairs (AraBERT on ArSarcasm-v2; MARBERT, MARBERTv2 and CAMeLBERT-DA on ArSAS). On validation it was significant in seven of eight, with increases of 2.0–3.3 points. Across both partitions, losing the negation word changes the predicted class of at most one additional tweet in thirty. These models evidently do not rest their

Edit	Model	ArSarcasm-v2 RAW	ArSarcasm-v2 STD	ArSAS RAW	ArSAS STD
Tatweel	MARBERT	17.3	0.0	9.7	0.0
	MARBERTv2	21.6	0.0	11.3	0.0
	AraBERT-tw	20.0	0.0	11.0	0.0
	CAMeL-DA	19.4	0.0	11.3	0.0
	XLM-R	13.4	0.0	9.6	0.0
Elongation ×2	MARBERT	10.2	0.2	10.3	0.1
	MARBERTv2	11.1	0.3	11.3	0.2
	AraBERT-tw	15.4	0.3	9.4	0.1
	CAMeL-DA	14.7	0.3	15.3	0.1
	XLM-R	16.2	0.4	12.3	0.2
Folding	MARBERT	2.7	2.8	2.6	2.6
	MARBERTv2	2.2	2.2	2.2	2.2
	AraBERT-tw	3.6	3.6	2.6	2.6
	CAMeL-DA	6.2	6.3	3.7	3.6
	XLM-R	8.1	8.1	5.2	5.2

Table 6: Test-set flip rates (%) under dense edits, without (RAW) and with STD normalization of clean and edited text.

decisions on the negation word alone.

Elongating the negation word (لاااي) increases flip rates significantly in only three of ten test pairs, and in two of ten on validation. The annotators read this edit very differently from tatweel. Annotator A judged the sentiment changed in 26 of 30 items, and annotator B in 16 (with 13 marked unsure). After adjudication, none of the 30 was judged to keep its sentiment (§5). An elongated لاااي is an emphatic refusal, and a model that changes its prediction may be reading it correctly.

8 Does normalization help?

Table 6 compares STD with RAW on the test sets. STD removes tatweel and collapses letter runs, and so reverses exactly the edits it targets. It brings flip rates under tatweel to zero and under dense elongation to at most 0.4% (H4: 70 of 70 tests on test, as on validation). It does not touch folding or typos.

The more relevant question is what STD costs on unedited text, where natural letter runs may carry emphasis. The cost is negligible. On the clean test tweets used in the tatweel conditions (2,948 and 3,898), Macro-F1 under STD differs from RAW by at most 0.35 points for every model, and it is higher under STD for six of the ten model–dataset pairs.

On validation we also compared the two conservative normalizers from our earlier work (Appendix Table 14). They remove most tatweel effects, leaving 0.5–1.0% of flips because they keep tatweel inside the words they treat as negation markers. They do not undo elongated long vowels, which their repetition rule excludes. The negation guard that distinguishes NASN2 from NO_CTX changes nothing that matters. Across 110 paired contrasts the two differ by at most 0.2 points. None favours NASN2, and five favour NO_CTX by fractions of a point. Under the preregistered rule, the claim that negation-aware normalization improves robustness is not supported (H5).

	ArSarcasm-v2	ArSAS
Tweets (train + val.)	12,546	15,704
with tatweel	571 (4.6%)	502 (3.2%)
with a letter run ≥3	498 (4.0%)	436 (2.8%)
of which a long vowel	300 (2.4%)	294 (1.9%)
tatweel on a negation word	1	1
letter run on a negation word	5	3

Table 7: Naturally occurring noise in the training and validation partitions.

9 Natural noise

Controlled edits show what can happen; natural noise shows what does (Table 7). The ArSarcasm-v2 test set is noisier still: 219 of its 3,000 tweets (7.3%) contain a tatweel and 339 contain a letter run. For these tweets we compare the prediction for the tweet as written with the prediction after STD cleaning (Table 8).

Cleaning natural noise moves many predictions. On ArSarcasm-v2 test, removing tatweel changes 7–12% of them and collapsing letter runs 16–20%. On ArSAS test the figures are 1–4% and 4–9%.

For letter runs on ArSarcasm-v2 the change is also an improvement. Macro-F1 is higher on the cleaned tweets for all five models, by 3.9 to 7.8 points, and every interval lies above zero. For natural tatweel on ArSarcasm-v2, on the smaller ArSAS slices (n = 89 and 128) and on the validation slices (n = 42–123), the differences are not significant. Collapsing a run removes emphasis that human readers notice (§5). For these classifiers, however, the emphasized form appears to be harder to read than the plain word.

10 Discussion

The tokenizer view and the prediction view tell different stories. A WordPiece model receives an `[UNK]` in place of every tatweel word, which sounds catastrophic. Yet the prediction usually survives, because sentiment in a tweet rarely depends on a single word. Where the two views agree is in the cost of the predictions that do change. When an Arabic WordPiece model loses a word and changes

Model	ArSarcasm-v2		ArSAS	
	Tatweel (n=219)	Letter run $\geq$3 (n=339)	Tatweel (n=128)	Letter run $\geq$3 (n=89)
MARBERT	8.8 [5.6, 12.3]	18.8 [15.5, 22.1]	3.1 [1.0, 5.8]	4.9 [2.2, 7.9]
MARBERTv2	11.7 [8.1, 15.5]	18.1 [14.7, 21.5]	4.2 [1.6, 7.2]	9.0 [4.9, 13.9]
AraBERT-tw	8.1 [5.0, 11.3]	16.1 [13.1, 19.4]	2.3 [1.0, 3.9]	4.1 [1.1, 7.5]
CAMeL-DA	8.5 [5.6, 11.6]	19.6 [16.4, 22.9]	2.6 [0.8, 4.9]	6.0 [2.6, 9.7]
XLM-R	7.3 [4.9, 9.9]	17.5 [14.6, 20.5]	1.3 [0.3, 2.6]	5.2 [2.6, 8.2]

Table 8: Test tweets that already contain noise: flip rate (%) between the tweet as written and its STD-cleaned form, with 95% intervals.

its prediction, the change costs more Macro-F1 than a change made by XLM-R, which merely fragments the word. On ArSarcasm-v2 it costs about 2.5 times as much.

The families also call for different readings. Tatweel is pure typography. Annotators saw no change in 90 of 90 items, so every prediction change under tatweel is an error, and a trivial fix removes it. Elongation is not noise in the same sense: readers took it as a change of intensity in most items, and a model that responds to it is not necessarily wrong. Benchmarks that treat all orthographic variation as a meaning-preserving perturbation conflate these cases.

The negation results are a caution against overinterpreting tokenizer audits. Deleting the negation word from a WordPiece model's input sounds as if it should invert many predictions. In practice it adds only a few, and the effect did not fully replicate on the test sets. The models appear to read sentiment from the rest of the tweet as well.

The practical recommendation is straightforward. Anyone deploying an Arabic WordPiece model on social media text should remove tatweel before tokenization. The cost on clean text stayed within 0.35 Macro-F1 points for every model, and on ArSarcasm-v2 cleaning natural letter runs improved accuracy as well. Collapsing letter runs does remove emphasis that readers perceive, so it should be a deliberate choice. Negation-aware exceptions to these rules did not help in our experiments.

Limitations

The edits are rule-based and applied one family at a time, whereas real noise co-occurs. The negation lexicon is finite and leaves out ما. The human validation covers one dataset and 30 items per condition. The natural-noise slices are small, and their accuracy effects are significant only for letter runs on ArSarcasm-v2. Both datasets consist of tweets, and other genres may behave differently. We did not evaluate instruction-tuned LLMs, whose byte-level tokenizers resemble XLM-R's but whose behaviour under these edits remains to be measured.

Ethics statement

We use public research datasets under their licenses and release only tweet IDs, hashes, labels and aggregate results, never tweet text. Two expert annotators and a third expert adjudicator labelled the human-validation sample.

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A Hypotheses on both partitions

Table 9 counts the supported tests for each preregistered hypothesis on validation and test. H5 compares two normalizers that the test protocol did not include. Per-test estimates and intervals are in the released bundles.

	Validation		Test	
Hypothesis	ArSarcasm-v2	ArSAS	ArSarcasm-v2	ArSAS
H1	15/15	15/15	15/15	15/15
H2	4/4	2/4	4/4	3/4
H3	3/4	4/4	1/4	3/4
H4	35/35	35/35	35/35	35/35
H5	0/55	0/55	–	–

Table 9: Supported tests per preregistered hypothesis (95% interval excludes zero in the stated direction).

B Test results by condition

The two tables on this page list the test flip rate for every condition and model.

C Validation (development) results

The validation results were obtained during development, before the study was frozen. They agree with the test results in direction for every hypothesis.

D Reproducibility

Code, configurations and text-free artefacts are released with the paper. The artefacts include perturbation logs, split manifests, predictions keyed by text hash, bootstrap results, the preregistration with its test amendment, and the freeze tag. Every table in this paper is generated from the released analysis bundles by a script; no number was entered by hand. Training is deterministic for a fixed seed, GPU type and software stack, and checkpoint hashes are published so that regenerated weights can be verified.

Condition	MARBERT	MARBERTv2	AraBERT-tw	CAMeL-DA	XLM-R
Tatweel, 1 word	7.0	9.3	8.4	8.2	7.0
Tatweel, 30% of words	17.3	21.6	20.0	19.4	13.4
Tatweel on negation cue	5.5	8.0	8.7	5.5	7.1
Elongation ×2, 1 word	4.2	5.0	6.6	5.8	7.8
Elongation ×4, 1 word	6.0	6.5	9.4	6.6	10.5
Elongation ×2, 30% of words	10.2	11.1	15.4	14.7	16.2
Elongation ×2 on negation cue	3.9	4.3	6.2	3.6	5.9
Folding, 1 word	1.2	0.9	2.2	3.3	4.3
Folding, 30% of words	2.7	2.2	3.6	6.2	8.1
Typo (control), 1 word	6.8	8.0	7.7	9.0	9.0
Typo (control), 30% of words	15.5	16.8	18.8	19.7	19.1

Table 10: ArSarcasm-v2 test: flip rate (%) for every condition, RAW inputs.

Condition	MARBERT	MARBERTv2	AraBERT-tw	CAMeL-DA	XLM-R
Tatweel, 1 word	3.0	3.9	3.6	3.9	3.4
Tatweel, 30% of words	9.7	11.3	11.0	11.3	9.6
Tatweel on negation cue	3.4	4.5	3.8	4.4	4.7
Elongation ×2, 1 word	3.2	3.9	2.9	4.0	3.6
Elongation ×4, 1 word	4.4	5.4	4.3	4.2	5.3
Elongation ×2, 30% of words	10.3	11.3	9.4	15.3	12.3
Elongation ×2 on negation cue	2.2	2.8	2.7	3.3	4.2
Folding, 1 word	1.1	0.9	1.5	2.0	2.5
Folding, 30% of words	2.6	2.2	2.6	3.7	5.2
Typo (control), 1 word	4.5	4.7	4.1	5.4	4.5
Typo (control), 30% of words	17.4	16.2	16.8	18.6	17.5

Table 11: ArSAS test: flip rate (%) for every condition, RAW inputs.

Condition	MARBERT	MARBERTv2	AraBERT-tw	CAMeL-DA	XLM-R
Tatweel, 1 word	5.2	6.0	6.6	5.1	5.0
Tatweel, 30% of words	15.4	17.5	15.9	14.9	11.8
Tatweel on negation cue	6.9	7.9	8.4	4.7	5.4
Elongation ×2, 1 word	4.1	4.3	4.9	5.5	6.8
Elongation ×4, 1 word	5.6	6.1	7.7	5.8	10.2
Elongation ×2, 30% of words	12.9	12.2	16.1	16.5	18.6
Elongation ×2 on negation cue	5.1	4.8	7.0	4.5	7.5
Folding, 1 word	1.4	1.6	2.1	2.8	3.6
Folding, 30% of words	3.2	3.1	4.0	5.4	6.7
Typo (control), 1 word	5.6	6.5	5.8	5.8	6.8
Typo (control), 30% of words	16.7	16.1	16.1	17.0	20.6

Table 12: ArSarcasm-v2 validation: flip rate (%) for every condition, RAW inputs.

Condition	MARBERT	MARBERTv2	AraBERT-tw	CAMeL-DA	XLM-R
Tatweel, 1 word	3.0	3.6	3.9	3.6	3.4
Tatweel, 30% of words	8.6	10.9	9.9	10.9	9.2
Tatweel on negation cue	4.2	5.9	5.5	4.3	5.1
Elongation ×2, 1 word	2.5	3.4	2.8	3.8	3.8
Elongation ×4, 1 word	3.9	4.9	3.8	4.3	5.4
Elongation ×2, 30% of words	9.7	11.1	8.9	15.7	11.7
Elongation ×2 on negation cue	1.9	2.4	2.8	2.3	4.9
Folding, 1 word	0.9	0.9	1.2	2.3	2.4
Folding, 30% of words	2.3	2.2	2.3	4.4	5.0
Typo (control), 1 word	4.1	4.5	4.2	5.4	4.4
Typo (control), 30% of words	16.7	16.0	18.0	19.3	18.3

Table 13: ArSAS validation: flip rate (%) for every condition, RAW inputs.

Edit	Model	ArSarcasm-v2 RAW	STD	NO_CTX	NASN2	ArSAS RAW	STD	NO_CTX	NASN2
Tatweel	MARBERT	15.4	0.0	0.8	0.8	8.6	0.0	0.6	0.6
	MARBERTv2	17.5	0.0	0.9	0.9	10.9	0.0	0.9	0.9
	AraBERT-tw	15.9	0.0	1.0	1.0	9.9	0.0	0.9	0.9
	CAMeL-DA	14.9	0.0	0.7	0.7	10.9	0.0	0.5	0.5
	XLM-R	11.8	0.0	0.8	0.8	9.2	0.0	0.7	0.7
Elongation $\times 2$	MARBERT	12.9	0.1	12.2	12.2	9.7	0.0	9.7	9.6
	MARBERTv2	12.2	0.2	11.6	11.6	11.1	0.1	10.3	10.3
	AraBERT-tw	16.1	0.2	15.4	15.5	8.9	0.2	8.7	8.7
	CAMeL-DA	16.5	0.2	15.8	15.9	15.7	0.1	15.1	15.0
	XLM-R	18.6	0.2	17.4	17.5	11.7	0.2	11.6	11.5
Folding	MARBERT	3.2	3.1	3.1	3.1	2.3	2.3	2.4	2.4
	MARBERTv2	3.1	3.1	3.1	3.1	2.2	2.3	2.2	2.2
	AraBERT-tw	4.0	4.0	3.9	3.9	2.3	2.3	2.3	2.3
	CAMeL-DA	5.4	5.3	5.4	5.4	4.4	4.3	4.3	4.3
	XLM-R	6.7	6.5	6.6	6.6	5.0	5.1	5.0	5.0

Table 14: Validation flip rates (%) under dense edits for all four preprocessing defenses, including the conservative normalizers NO_CTX and NASN2.

	ArSarcasm-v2 Tatweel	Elong. $\times 2$	Typo	ArSAS Tatweel	Elong. $\times 2$	Typo
MARBERT	5.9 [4.6, 7.2]	4.4 [3.4, 5.3]	7.5 [6.2, 8.7]	3.5 [2.5, 4.6]	4.6 [3.6, 5.6]	10.9 [9.4, 12.3]
MARBERTv2	6.2 [4.8, 7.7]	2.9 [1.9, 3.9]	5.4 [4.1, 6.7]	4.0 [2.8, 5.2]	4.3 [3.2, 5.5]	7.3 [5.8, 8.8]
AraBERT-tw	5.0 [3.6, 6.3]	3.7 [2.4, 4.9]	5.4 [3.9, 6.8]	3.7 [2.6, 4.9]	3.3 [2.3, 4.3]	9.8 [8.2, 11.4]
CAMeL-DA	4.3 [2.9, 5.6]	4.2 [3.0, 5.5]	5.4 [4.0, 6.9]	3.7 [2.6, 4.8]	6.1 [4.7, 7.5]	9.2 [7.6, 10.7]
XLM-R	1.5 [0.6, 2.5]	5.5 [4.4, 6.7]	7.5 [6.1, 8.9]	1.4 [0.5, 2.4]	3.6 [2.6, 4.8]	7.1 [5.6, 8.6]

Table 15: Validation Macro-F1 drop in points under dense edits, with 95% intervals.